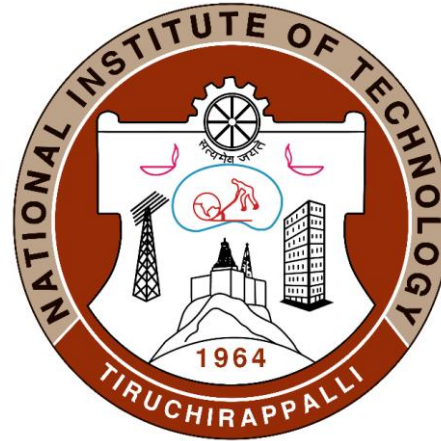


Introduction to Polymers as an Engineering Material



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Lecture - 5

High Temperature polymers

High Temp polymer

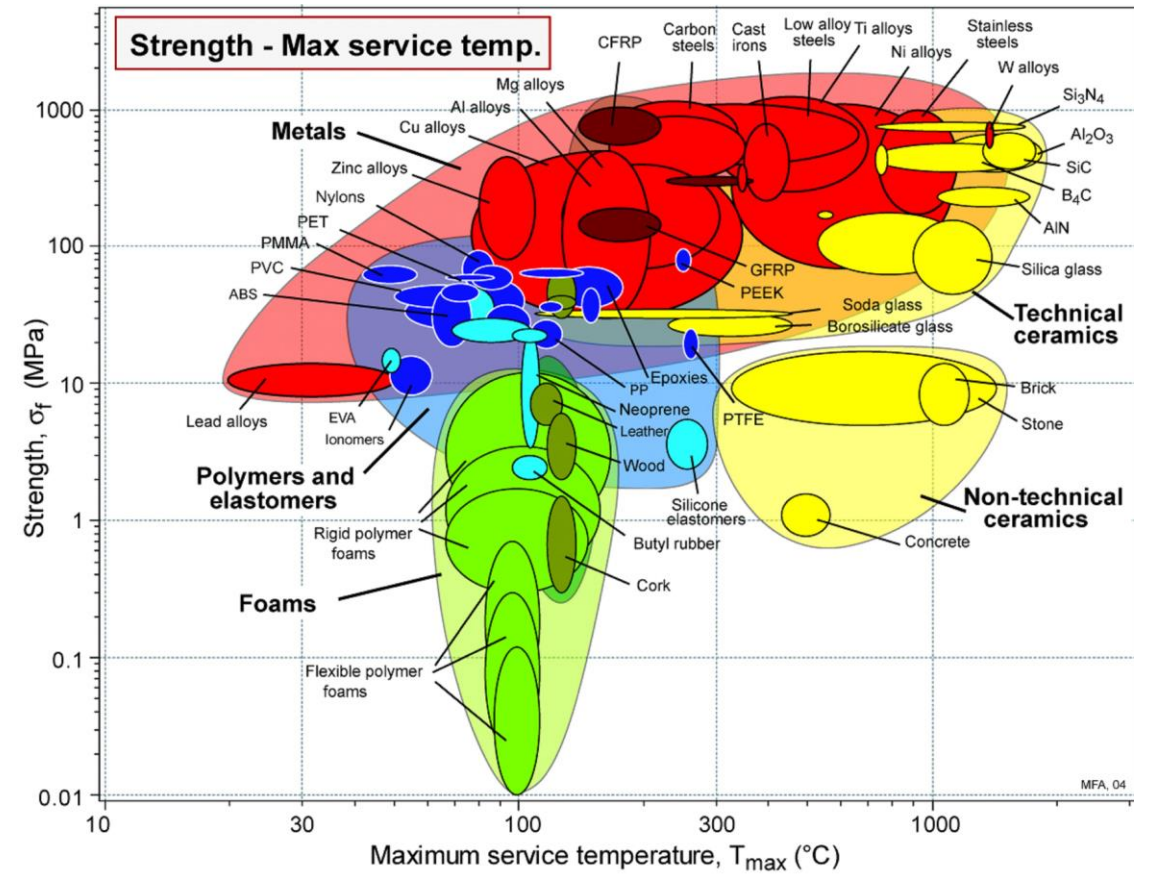
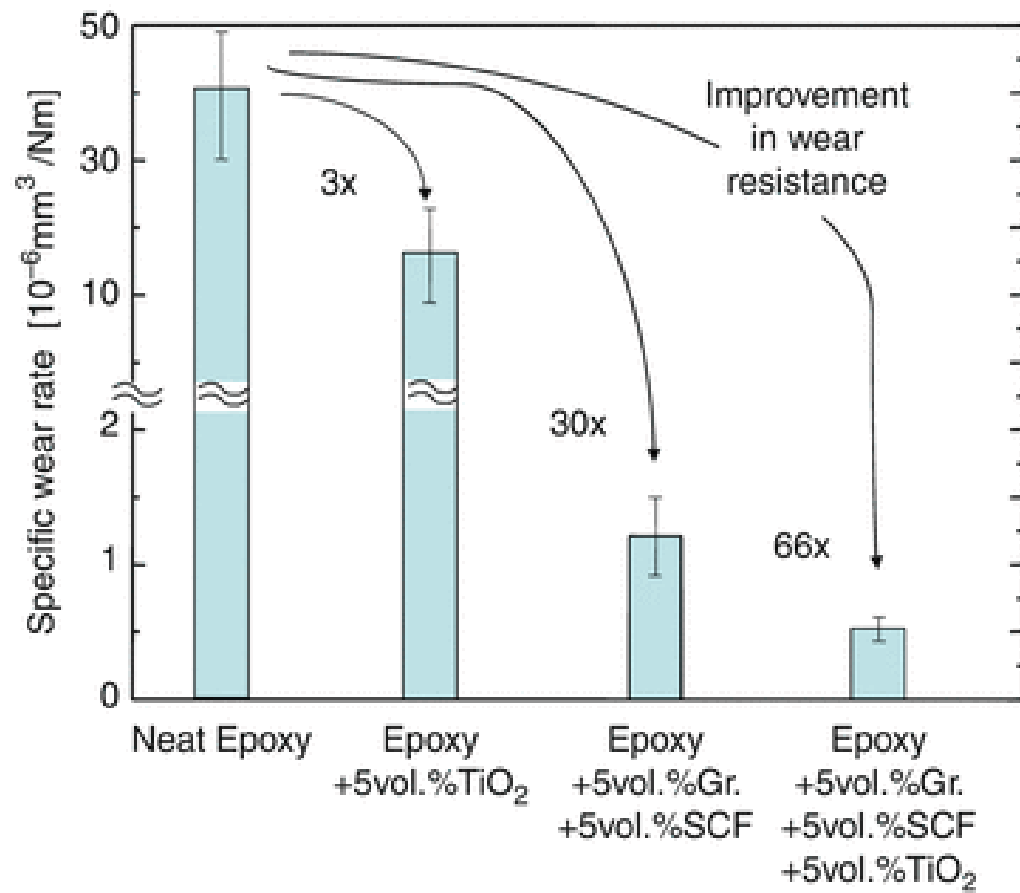
Plastics which can retain useful properties for thousands of hours at 230 °C (503 K) or above or for hundreds of hours at 300 °C (573 K), or some minutes at 540 °C (813 K) or for some seconds up to 760 °C (1033 K) can be regarded as high temperature polymers.

These temps may be encountered during manufacturing or use of a give product.

These materials can withstand high temperature not only without melting but also without losing their strength

For instance, construction of a spacecraft, a satellite or a supersonic aircraft needs materials that can withstand high surface temp, have low density and posses high dimensional stability

In case of electrical engineering, motors can function more efficiently at higher temperature provided the material for insulation with greater thermal stability is available



They should possess high thermo-oxidative stability

According to NASA, high temperature plastics are those which can withstand service temperature in excess of 200 °C (473 K) in air up to 50000 hrs without appreciable deterioration of physical properties.

High Temperature polymers can be grouped as

- (i) Polyamides of various types which include condensation polyimide (CPI), addition polyimides (API) Acetylene terminated polyimide (ATPI), Polyamideimide (PAI), Polyester imide (PEI) and bismaleimide (BMI)
- (ii) Polyarylenes esters, Polyphenylene sulphide (PPS) and Liquid crystal polymers

Basic Principles

There are several ways by which the heat stability of polymers can be increased

- (1) Strength of the bonds which join different groups in the main polymer chain, i.e., in the polymer backbone. For instance, a polymer chain with Si-O bonds would be more stable than one with C-C bonds, because Si-O bond has higher bond energy than C-C bonds. That is, replacement of a carbon atom in the main polymer chain by inorganic atoms or grouping in one way of producing structures with greater heat resistance

Table Bond energies of different bonds

Bond Type	Bond Energy, kJ/mole
C-C	347
Si-O	373
Si-N	462
B-N	437
B-O	500

2. Incorporation of aromatic ring structures in the polymer backbone.

The aromatic rings i.e. the phenylene rings are stiffer and more resistant to deformation and produce intrinsically stronger bonds and hence high resistance to heat

Ex. This shows that substitution of p-phenylene group for six chain CH₂ groups causes a marked rise in the melting point of the polymer

Similarly, a polymer chain, in which different groups are joined with two strands, has been found to be stronger than the single stranded polymer. A double stranded polymer (also called ladder polymers) structure is obtained by fusing different rings in a linear fashion

3. Incorporation of highly stable and rigid heterocyclic ring system in the polymer chain also increases the heat resistance of polymers.

In fact it has been observed that the structures formed by incorporating heterocyclic rings in the backbone are generally not meltable and decompose only above 500 °C.

4. Slight changes in the structures of the monomers can have a marked influence on the physical properties of the polymers so obtained. It has been observed that para-and meta structures which are relatively symmetric have higher inter chain attraction and can crystallize better than ortho-structures

(5) The addition of short and bulky side chains at regular intervals is another method of increasing heat resistance. The stereo-regularity causes crystallization. As a consequence the side chains tend to interlock mechanically and thereby reduce the tendency of the molecules to slide over one another when heat energy is applied.

(6) Polymers containing halogens (e.g. PTFE) can be used for continuous service at temperatures of about 150 C (423 K) or more.

Examples of high temperature polymers are : aromatic polyamides (Aramid), Polyimides (PI)etc